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(7th Joint Command and Staff Course)

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Thesis Title:

Stronger Together: How can the Defence Forces Leverage Defence Standardisation to Enhance its Interoperability with Partners?

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STUDENT DECLARATION

1. I certify that this thesis does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any university; and that to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in the text.
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ABSTRACT

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Defence standardisation plays an important role in modern military operations, with its primary goals centred on enhancing interoperability, improving resource management and boosting operational effectiveness. This thesis analyses the current state of defence standardisation within the Irish Defence Forces and examines how defence standardisation can be leveraged to build military capabilities that are interoperable with regional partner nations.

This thesis utilises DOTMLPF-I and DOTmLPF-P as frameworks from which to examine three main questions. Firstly, what is defence standardisation, what are its aims and potential impacts on the Defence Forces? Secondly, how does the current defence standardisation landscape within the Defence Forces function? Thirdly, how do European partners manage defence standardisation, and what lessons can the Defence Forces adopt to enhance governance, implementation, and engagement?

Through the examination of the current state of defence standardisation it identifies three key areas that the Defence Forces can work on to improve standardisation implementation and its end state of interoperability. These three areas include weak governance, ineffective implementation and inconsistent engagement. This thesis examines options to address this through an analysis of partner's governance structures and mechanisms, implementation planning and management and standardisation engagement and understanding.

Finally, it suggests a tailored solution for the Defence Forces to address the areas outlined above based on the experience and knowledge of the Defence Forces European partners. The thesis concludes by identifying standardisation as a key tool that allows for the development of a firm foundation from which to build military capabilities that are interoperable with its European partners.

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GLOSARY OF TERMS

A2IND – French Defence Interoperability and Standardisation Support Agency

AAP – Allied Administrative Publication

AJP – Allied Joint Publication

ANSI – American National Standardisation Institute

ATP – Allied Tactical Publication

Bi-SC – Bilateral Strategic Command

CC&CS – Capability Codes and Capability Statements

COPD – Comprehensive Operational Planning Directive

CS – Committee for Standardisation

CSDP – Common Security and Defence Policy

DFDM – Defence Forces Doctrinal Manual

DFHQ – Defence Forces Headquarters

DStan – United Kingdom Defence Standardisation

DSA – Defence Safety Authority

DTA – Delegated Tasking Authority

EDA – European Defence Agency

EDSC – European Defence Standardisation Committee

EDSMG – European Defence Standardisation Management Group

EDSTAR – European Defence Standardisation Reference System

EDTIB – European Defence Technical Industrial Base

EUMS – European Union Military Staff

IRLJP – Irish Joint Publication

ISO – International Organisation for Standardisation

ITPP – Individually Tailored Partnership Programme

JIC – Just in Case

JIT – Just in Time

JSP – Joint Service Publication

MC – Military Committee

MOE – Measure of Effectiveness

MOP – Measure of Performance

MTT – Mobile Training Team

NAC – North Atlantic Council

NATO – North Atlantic Treaty Organisation

NDPP – NATO Defence Planning Process

NSAI – National Standardisation Authority of Ireland

NSDD – NATO Standardisation Document Database

NSO – NATO Standardisation Office

OCC – Operational Capabilities Concept

OECD – Organisation for Economic Cooperation and Development

OUP – Oxford University Press

PfP – Partnership for Peace

PME – Professional Military Education

SME – Subject Matter Expert

SMG – Standardisation Management Group

STAMIS – Standardisation Management Information System

SWOT – Strengths, weaknesses, opportunities, threats

TA – Tasking Authority

TI – Training Instruction

UK MoD – United Kingdom Ministry of Defence

US DoD – United States of America Department of Defence

INTRODUCTION

Introduction.

The implementation of the changes outlined in the Report of the Commission on the Defence Forces 2022 offers a crucial opportunity for the Defence Forces to enhance military capabilities for national defence and contributions to international peace and security. Central to this transformation is Defence Standardisation with regional partners, particularly through the NATO Partnership for Peace (PfP) programme, which Ireland has engaged with for over 25 years. Despite this the Defence Forces have not fully leveraged standards, essential for achieving interoperability with regional partners.

This thesis will examine barriers to effective engagement with defence standardisation bodies, including weak governance, ineffective standardisation implementation and inconsistent stakeholder engagement. It proposes strategies to improve structures, implementation and engagement enhancing interoperability and international collaboration with international partners in a complex global security landscape.

As defence reforms progress, Defence Standardisation with our regional partners offers the Defence Forces a cost effective foundation from which to effectively develop military capabilities for national defence and to contribute to international peace and security. The PfP programme provides a framework for the adoption of NATO standards, which aligns with national defence policy, however, challenges within governance, implementation and engagement persist.

To address this, the research will assess current standardisation management, identifying its strengths, weaknesses, opportunities and threats. Developing an understanding of what current standardisation gaps are and exploring options to address them through regional collaboration. This this will propose a tailor made standardisation solution, based on regional best practice, to meet the Defence Forces governance, implementation and engagement requirements. This solution would improve governance, implementation and allow the Defence Forces to leverage regional expertise to create cost-effective, interoperable capabilities aligned to best practice regional standards.

Literature Review.

Standardisation is a cross-disciplinary field spanning engineering, procurement and defence. It is critical for enhancing interoperability within in military alliances and partnerships. The body of literature addressing standardisation can be categorised into two broad schools of thought, which are explored further below. These schools of thought present differing perspectives on the central issues and their insights have been integrated in order to guide the research of this thesis. These schools of thoughts being the technical-optimistic, which views standardisation as a driver of efficiency and the socio-political, which highlights its contested nature. These perspectives inform the thesis's examination of the role of defence standardisation within interoperability.

The technical-optimistic school, represented by a body of work on standardisation from Perry, Verman, Spivak and Brenner, which frames standardisation as a systematic tool for progress, emphasizing technical and procedural benefits across industries.

Perry's (1955) work is reflective of standardisation's history from ancient times to the industrial era, focusing on its role in technological and economic advancement. Writing post-World War II from a US perspective, he highlights national standards bodies like the National Bureau of Standards and international efforts for compatibility. Perry views standardisation as a catalyst for manufacturing efficiency and global trade, citing examples like interchangeable parts. His optimistic narrative, however, overlooks political barriers.

Verman's (1973) work establishes standardisation as a distinct discipline, advocating a global, systematic approach to uniformity in products and processes. His foundational work, influential in Spivak and Brenner's scholarship, emphasizes procedural rigor to optimise systems. While acknowledging challenges like divergent national standards, Verman's focus remains on technical solutions.

Spivak and Brenner (2001) offer a practitioner-oriented guide, covering standardisation's history, global implementation and real-world applications. They discuss ISO standards' role in trade and note potential barriers, providing a more nuanced technical perspective. Their case studies illustrate standardisation's efficiency in industries, including defence.

In military contexts, this school suggests standardisation enhances interoperability through compatible equipment and protocols. Perry's historical insights highlight standardised

weaponry, while Verman and Spivak and Brenner advocate frameworks for defence logistics, like uniforms, ammunition and communication systems. Yet, this perspective often underestimates political complexities.

The socio-political school of thought which is favoured by Ford and Gould (2019). This school of thought views standardisation as a contested process shaped by competing interests, particularly in military settings.

Ford and Gould's (2019) work analyses NATO's 1970's standardisation efforts during the Cold War. It emphasizes its importance for conventional defence against the Warsaw Pact. Using Actor Network Theory, they reveal how national priorities, industrial concerns and political rivalries hindered progress. Their work underscores standardisation's socio-political nature, contrasting with the technical-optimistic view.

This school highlights barriers to military interoperability, such as nations protecting domestic industries, which technical solutions alone cannot resolve. It complements the thesis by exposing practical challenges in alliances like NATO.

The technical-optimistic school of thought provides tools for designing interoperable military systems, such as standardised equipment, while the socio-political school reveals implementation barriers, like political resistance. For example, a standardised rifle may be technically feasible but contentious if it favours one nation's industry. The former offers procedural frameworks, while the latter stresses diplomatic coordination, such as NATO-led negotiations.

In defence, successful standardisation balances technical rigor with political strategy. The technical-optimistic school informs standards creation, while the socio-political school underscores the need for mechanisms like joint procurement to align interests.

An examination of the literature reveals two schools of thought: technical-optimistic, celebrating standardisation's efficiency and favoured by Perry, Verman, Spivak and Brenner and socio-political, highlighting its challenges, favoured by Ford and Gould. For defence interoperability, integrating these perspectives ensures robust standards and strategic alignment, addressing both technical and political dimensions.

Additional Sources.

In addition to academic texts, it is important also to refer to military sources in order to understand the ways in which standardisation drives interoperability. Military sources of importance include NATO's Interoperability Policy (2009) and Standardisation Policy (2016), which emphasise that "Standardisation will continue to be a key element in achieving and maintaining interoperability" (NATO, 2009, p. 2-2). These policies apply to all NATO Allies and encourage Partner Nations to adopt them nationally. Complementing this, the European Defence Agency's Standardisation Policy (2024) focuses on capability development through standardisation, promoting awareness among Member States and Industry and harmonising positions for a unified approach. *Allied Administrative Publication (AAP) – 03* (NATO, 2022a) outlines a seven-phase standardisation process for managing standardisation. It emphasises commitment to implementation, timeliness, transparency, and consistency in developing standardisation documents. The NATO public webpage provides information on the Partnership for Peace (PfP) programme (NATO, 2022b). It offers access to interoperability and standardisation tools like the Partner Cooperation Menu, Planning and Review Process and the Operational Capabilities Concept (OCC).

Research Lacunae.

Having analysed the available literature on standardisation, it is clear that a number of lacunae exist. First of all there is a dearth of research in respect of how standardisation can be applied within smaller militaries such as the Defence Forces. The available research covers industry in depth but overlooks its military application. While some research is available in respect of NATO members, this work does not provide insight into the challenges faced by partner nations when implementing standardisation programmes. Other available literature such as NATO policy documents emphasise interoperability, however, lack specific solutions to challenges within governance, implementation and engagement. Further to this, there is a significant gap in Defence Forces research into how the organisation can leverage NATO standardisation through the PfP programme, despite 25 years of participation. This dearth of research further ignores barriers to standardisation such as weak governance, ineffective implementation and inconsistent stakeholder engagement, which limit interoperability with partners. This gap is key as it acts as a barrier to national defence policy implantation, which recognises NATO as the standard that the Defence Forces should meet. This research seeks to investigate these challenges and suggest a toiled solution to meet the Defence Forces requirements for

standardisation governance, implementation and engagement in order to enhance interoperability and international collaboration with international partners in a complex global security landscape.

Research Questions.

This thesis will address the following question: How the Defence Forces can leverage defence standardisation to enhance interoperability with partners?

1. What is defence standardisation, what are its aims and potential impacts on the Defence Forces?
2. How does the current defence standardisation landscape within the Defence Forces function?
3. How do European partners manage defence standardisation, and what lessons can the Defence Forces adopt to enhance governance, implementation and engagement?

These research questions address the main aspects of the proposed solution, including the developing of a clear understanding of defence standardisation and its potential impacts, the current defence standardisation landscape within the Defence Forces and what gaps exist and how our European partners manage standardisation and how we can implement best practice defence management structures and mechanisms within the Defence Forces. These questions will inform the structure of this thesis, with the questions forming the three main chapters.

Methodology.

The methodology of this thesis employs qualitative analysis utilising both thematic literature reviews, historical and comparative approaches in its research concept. Developing an understanding of the existing literature will require qualitative analysis of current defence standardisation publications and papers from National and International defence standardisation organisations. Furthermore, to understand current best practices in defence standardisation governance structures and mechanisms analysis of policy publications and best practice standardisation management models from our European partners will be necessary.

This is to establish what defence standardisation is, through a clear definition, its aims are and how its effective application would impact the Defence Forces utilising a DOTLMPF-I¹ versus SWOT analysis. Applying a SWOT analysis to the elements of DOTMLPF-I and DOTmLPF-P provides a granular examination of how standardisation is applied within the organisation and allows for the development of strategic insights in respect of strengths, weaknesses, opportunities and threats across the overarching system. The combination of these frameworks providing more than the sum of their individual parts.

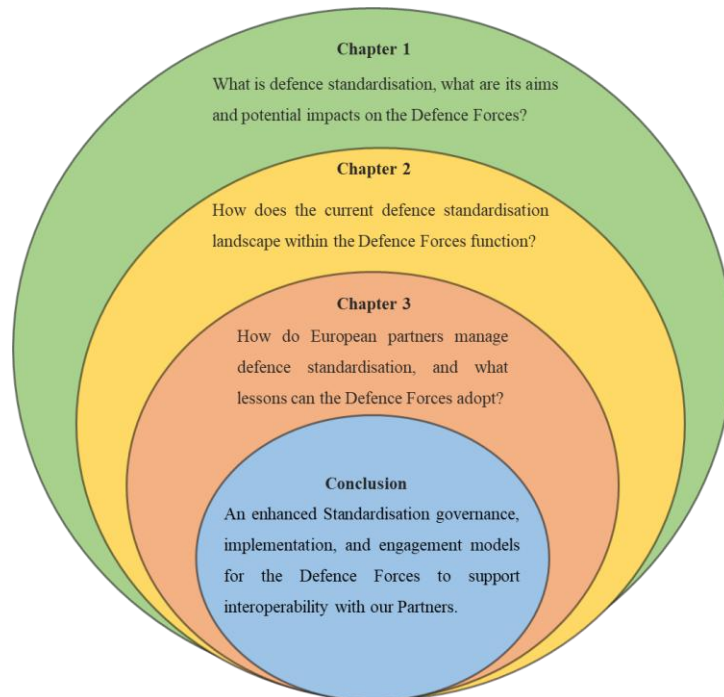


Figure 1: Research Framework

The thesis will approach the research within a framework that links the research questions, building towards its conclusion of how to enhance Standardisation governance, implementation, and engagement models for the Defence Forces to support interoperability with our Partners (*As per figure 1*).

Thesis Outline.

Chapter 1 will examine key standardisation definition, outline the aims of standardisation, examine the historical development and importance of standardisation across a number fields including engineering, commercial and the military, and it will finally analyse the potential

¹ Doctrine, Organisation, Training, Materiel, Leadership, Personnel, Facilities and Interoperability (DOTMLPF-I) is a framework utilised by NATO to understand the various aspects that make up a military capability.

impact of standardisation on the Defence Forces, through a DOTMLPF-I versus SWOT analysis.

Chapter 2 will analyse the current standardisation landscape within the Defence Forces through a DOTmLPF-P versus SWOT analysis. This will examine how and where standardisation is implemented, identifying the strengths of the current landscape, its weaknesses, opportunities for enhancement and potential threats posed by remaining with the current defence standardisation landscape.

Chapter 3 will examine how standardisation is managed by our European Partners. Analysing the regional and national governance structures and mechanisms, key standardisation management publications and best practice standardisation management models in order to deliver a model suitable to meet the needs of the Irish Defence Forces.

The conclusion chapter outlines the findings of the research and suggests areas for further research.

CHAPTER ONE – UNDERSTANDING DEFENCE STANDARDISATION, ITS AIMS AND POTENTIAL IMPACTS.

1.1. Introduction.

This chapter, which is divided into three parts, will explore the definitions, goals of standards, standardisation and related topics, it will then examine the historical development and evolution of standards from a local to a global tool and finally it will analyse the benefits of defence standardisation to the Irish Defence Forces, with a focus on European defence standardisation actors.

Part 1 of this chapter will examine the key definitions within the discipline of standardisation, the aims of standardisation and will then examine the component parts that make up standardisation. This chapter will achieve this through an examination of how standards and standardisation are defined across a broad range of disciplines, focusing the military aspects and outlining and examining what the desired outcomes are for standardisation within them.

Part 2 will examine the historical emergence and development of standards and standardisation. I will achieve this by exploring the emergence of standardisation from its early historical development, through the past industrial revolution growth, developments within the Post-World War landscape to enhancements which have occurred during modern times. This historical examination will take a broad look at the global, regional and national actors operating within the standardisation community and will include perspectives from an engineering, commercial and military view point.

Part 3 will then examine the impact of standardisation on a small defence forces. To achieve this I will compare the strengths, weakness, opportunities and threats posed by the various effects of standardisation across the capability development framework of doctrine, organisation, training, materiel, leadership, personnel, facilities and interoperability. Concluding with a summary the finding of this analysis.

1.2. Definition, Goals and Components of Defence Standardisation.

From a literal perspective a standard is defined in the oxford dictionary as a rule, criterion or measure by which something can be judged or evaluated (OUP, 2022). Within the field of engineering, where standards provide key norms such as a unit of measure and quality, a

standard has been described as an agreed way of doing something (Spivak & Brenner, 2001). Building on this, civilian standardisation bodies have described a standard as a way of communicating, in the form of a technical specification, to avoid misunderstanding and that exist for our benefit (Danish Standardisation Foundation, 2015). However, from a military perspective, a standard has been described as rules, guidelines or characteristics aimed at the achievement of the optimum degree of order in a given context (NATO, 2022b). The common themes throughout all of these definitions being that a standard is a set of rules or guidelines that are used to communicate, evaluate and achieve consistency within a specific field. They serve as a reference point for quality, performance or procedures that minimise misunderstandings, enhance cooperation and lead to optimum outcomes. Standards provide the means by which to achieve standardisation. Standardisation itself has been described as the process of creating, issuing and implementing standards (Dept. of Business and Trade, 2025). Choi adds to this by describing it as the act of making a document a standard (Choi et al, 2010). Once again from a military perspective, NATO define standardisation as the development and implementation of procedures, designs and terminology to the level necessary for the interoperability (NATO, 2016). In effect standardisation is the process by which standards are implemented to achieve success.

The primary aim of standardisation is the reduction of complexity and cost while increasing efficiency and quality, through the development of universal, transparent and uniform means of communication, that remove barriers to cooperation, fostering interoperability.

As the complexity of society has exponentially increased standardisation provides a means to combat this through the simplification of practices and processes (Saunders, 1972). A key component of this simplification is the reduction of variation and the increase of interchangeability. This simplification must also be based upon the adoption of the best possible solutions to problems utilising the most up to date knowledge and technology (Verman, 1973). By its nature this practice promotes the development and implementation of simplified and ultimately best practice standards (EDA, 2024a). While reducing complexity will encourage standardisation, increasing economic efficiency will broaden the incentives to do so. To achieve this, the aim of the standards must be overall economy of effort. Delivering a balanced standard that addresses the economic benefits across the entire spectrum of the standards outcomes (Saunders, 1972). This should be examined taking the costs, human effort and resource conservation into consideration (Verman, 1973). Taking overall economy into

consideration will lead to industrial efficiency which will reduce overall costs (EDA, 2024a). Having dealt with the issues of complexity and cost we must be also tackle the issue of communication. All standards must aim to provide clear and uniformly understood direction on what the standard seeks to provide, utilising agreed symbols and codes (Saunders, 1972). This clear communication foster trust and promotes collaboration between standardisation stakeholders (EDA, 2024a). The final aim of standardisation is the fostering of cooperation between standardisation stakeholders, this is achieved through the removal of potential barriers to trade which may occur when standards are unnecessarily legislated. This can be mitigated by developing international standards through consensus of international expert groups (Saunders, 1972). This mitigation will maximise the standards convenience of use (Verma, 1973) and will promote the harmonisation of standards (EDA, 2024b) globally for the benefit of all standardisation stakeholders.

From a military perspective, defence standardisation enables smooth cooperation among national, multinational structures, forces and assets, which is essential for multi-national operations (NATO, 2022b). By implementing agreed defence standards, forces can operate together effectively, improving safety and efficiency in support of missions (NATO, 2014). It supports achieving, maintaining and enhancing interoperability among multinational forces, strengthening defence capabilities and operational efficiency (NATO, 2022b). Effectively implemented defence standardisation also facilitates efficient resource management by promoting the use of common procedures, designs and terminology, potentially reducing procurement costs and avoiding duplication of efforts (Blind, 2013). It is crucial for strengthening military capabilities and operational and logistical efficiency in multinational operations (Pernin et al, 2020). Effective standardisation implementation is crucial for achieving operational effectiveness, resource management and interoperability.

Defence standardisation plays an important role in modern military operations, with its primary goals centred on enhancing interoperability, improving resource management and boosting operational effectiveness. Interoperability creates the conditions to allow a multinational force to act together coherently, effectively and efficiently to achieve tactical, operational and strategic objectives (NATO, 2009). Interoperability is the key focus of standardisation efforts (NATO, 2016). Standardisation of operational, materiel and information interchange helps reduces associated risk and leads to increased interoperability (EDA, 2007). This promotes seamless cooperation during joint operations, reducing friction and improving overall mission

success through common doctrine, tactics and procedures. In terms of resource management, standardisation drives efficiency, quality and consistency across the defence sector by removing barriers, simplifying processes and procedures and enhancing the uniformity of materiel (Spivak & Brenner, 2001). By implementing uniform standards, procurement becomes more streamlined through the removal of duplication of effort, maintenance costs are reduced and the overall quality of equipment and systems is elevated (Verman, 1973). This approach not only saves valuable resources but also ensures that military personnel can rely on consistent, high-quality tools (EDA, 2007) and equipment regardless of their specific unit or location. Furthermore, defence standardisation significantly enhances operational effectiveness by reducing barriers (Spivak & Brenner, 2001) to cooperation between different military branches and international allies and by increasing efficiency of design, development, materiel acquisition and logistics support (US DoD, 1965). Common standards facilitate smoother communication, joint training exercises and shared logistics, ultimately leading to more cohesive and effective military operations (NATO, 2022b). In an unpredictable global environment standardisation provides a uniform means to communicate and cooperate. Within the defence profession this, uniform means that standardisation provides, can be a useful tool to build up national military capabilities by making use of the experience of others to build up a national foundational understanding of a given topic.

1.3. Historical Development of Defence Standardisation.

In order to better understand standardisation we need to understand where it came from and how it developed as a discipline. It has been argued that standards and the implementation of standardisation practices are a means to understand the world around us and are as old as civilisation and society itself (Perry, 1955). To demonstrate this and to establish where we are at today within the discipline I will outline this through a framework which will examine standardisations early origins, its post Industrialisation, Post-1945 and modern era developments.

Early origins.

Standardisation was certainly a concern for engineers as they began their work on the Egyptian pyramids. Modern analysis has shown that they were skilled users of uniform measurement, however, this measurement tended to be linked with the individual project of the society that it was part of (Perry, 1955). The uniformity of this early measurement was a key enabler of this

great project and others like it. In addition to engineering standards early commercial standardisation focused on the establishment of uniform weights and currency to safeguard against cheats and to promote transparency (Spivak & Brenner, 2001). By developing uniform standards early cultures promoted trade and cooperation. Standardisation was not just an engineering and commercial concept it also was an important consideration for early military forces. As militaries expanded the requirement to clothe, arm and support armies became a challenge. The requirements of mass demand within the Roman Army led to simplification and mass production of weapons and equipment (Coulston, 1998). While there is no evidence of a centrally designed and issued set of equipment there was a list of items that a soldier needed to have to perform their job.

Post Industrialisation.

The next major leap in standardisation took place following the industrial revolution. The advent of mass production and replaceable parts placed standardisation within the arena of the industrialist and the engineer (Verman, 1973). The concept of replaceable parts was also of interest to military profession and the concept of the standardised musket was developed in 1800. The design was required to meet increasing military requirements for a mass produced musket (Verman, 1973). Leading the way to military industrialisation. Two key units of measure also arose at this time, in France the metric system and in Britain, the imperial system (Perry, 1955). These measurement systems, while individually uniform, created competing standards to which things were build, measured or understood. This included the next step, the standardisation of the rail gauge. While a standard gauge of 4ft. 8½ in. or 1.44m which was identified within Britain in 1795 it was introduced at a time where there were multiple competing gauges. The driver of change ultimately came from the military when the US congress mandated the 4ft. 8½ in. to aid military logistics, which took until 1888 to implement (Spivak & Brenner, 2001).

Post-World Wars.

The increased industrialisation, technology and cooperation that took place during the globe spanning world wars led to ever increasing levels of standardisation. This increase in standardisation was led by a growing international standardisation movement made up of engineers and physicists linked with the electrification of society (Verman, 1973). Increased wartime industrial requirements also highlighted the high level of variety within tools, nuts and

bolts. This variation led to increasing costs as it became necessary to produce these items within the US and transport them to where they were required (Spivak & Brenner, 2001). The standardisation efforts that took place in response to this problem led to the creation of international standardisation bodies such as the International Standardisation Office (ISO) (OECD/ISO, 2016), regional standardisation bodies such as the NATO Standardisation Office (NSO) (NATO, 2024a) and national standardisation bodies such as the American National Standardisation Institute (ANSI) (Spivak & Brenner, 2001). This increasing standardisation effort led to global proliferation of standards led to a reduction in variation and costs while increasing interchangeability and cooperation potential, making standardisation a vital part of national planning (Verman, 1973).

Modern era.

The Cold War period was one of increased international relations competition between the East and West (Ford & Gould, 2019). The response on both sides was to standardise within their spheres. NATO began to establish standards for common tactical doctrines to further enhance alliance cooperation, which led to the concept of military interoperability (NATO, 2024a). The spread and implementation of these tactical standards has led to ever increasing levels of effective and efficient cooperation between forces operating in multinational environments. An example of this standardisation in practice is the development and implementation of standardised ammunition types across NATO members and other partner nations, enhancing nation's interoperability. Civil standardisation also continued and developed, within the automotive and aviation industries. The automotive industry implemented the metric system for motor parts and the aviation industry implemented a single set of standards for all aircraft parts to meet (Spivak & Brenner, 2001). These advances in standardisation within the automotive and aviation industries have improved safety and reduced costs. Finally, the increased access to information as a result of the computer age has meant that access to standards by organisations is easier than ever. Examples of this being the European Defence Agency's (EDA) European Defence Standards Reference System (EDSTAR) (EDA, 2025a) and NATO's NSO Website (NATO, 2025a). These websites provide easy access to key standardisation documents and information related to their implementation, simplifying access to them and encouraging broad implementation by allies, partners and industry.

The lesson to take from all of this is that standardisation has proven an effective response to the complex challenge of mass demand and mass production within large organisations or

projects. Whether this challenge relates to an engineering, commercial or military matters, standardisation provides a uniform solution to ever increasing complexity and cooperation. The aim of these solutions is to simplify practices, improves cooperation and reduces costs, these three areas being very important for military organisations which operate within a limited resource environment. It is because of these reasons that standardisation has been a fundamental part of militaries and the generation of military force throughout history.

1.4. Analysing the impact of Defence Standardisation on the Irish Defence Force.

In order to assess the utility of defence standardisation to the Irish Defence Forces this section will now explore its potential strengths, weaknesses, opportunities and threats (SWOT) analysis within a SWOT over DOTMLPF-I framework as per figure 2.

SWOT DOTMLPF-I	Strengths	Weaknesses	Opportunities	Threats
Doctrine	What are the benefits of doctrinal standardisation?	What challenges arise as a consequence of doctrinal standardisation?	What opportunities can doctrinal standardisation deliver?	What threats are posed as a result of defence standardisation?
Organisation	How can defence standardisation improve an organisation?	How may the organisation struggle to implement organisational standardisation?	What opportunities will organisational standardisation offer?	The threats posed by organisational standardisation?
Training	How can the implementation of training standards improve the organisation?	What are the likely shortfalls of training standardisation within the organisation?	What opportunities can be leveraged as a result of training standardisation?	What challenges are we likely to face during the implementation of training standardisation?
Materiel	How might the standardisation of materiel improve the organisation?	How might it potentially negatively impact the organisation?	What opportunities are likely to arise as a result of materiel standardisation?	What challenges may result through materiel standardisation?
Leadership	How can standardisation positively impact the PME of the organisations leadership?	How can standardisation create challenges within leadership PME?	What opportunities can be exploited as a result of leadership PME standardisation?	What potential threats may arise from standardisation of leadership PME?
Personnel	How can standardisation enhance personnel skills?	How can personnel skill standardisation negatively impact the organisation?	What opportunities are created as a result of personnel skill standardisation?	What threats are posed to the organisation as a result of personnel skill standardisation?
Facilities	How might facilities standardisation benefit the organisation?	What challenges are efforts to implement facilities standardisation likely to encounter?	What potential opportunities exist within facilities standardisation that can be exploited by the organisation?	How might facilities standardisation introduce new threats?
Interoperability	What benefits does interoperability offer to the organisation?	How might interoperability standardisation be challenged?	What opportunities does interoperability bring to an organisation?	What potential new threats are introduced as a result of the organisation seeking interoperability within our partners?

Figure 2 – DOTLPF-I – SWOT Analysis Matrix.

Doctrine.

The main strengths of standardisation are how it provides readymade doctrinal solutions that have been developed by experts and approved by a recognised international standardisation

development organisation such that can address gaps in national doctrinal architecture (Koops, 2023). This reduces the cost to create a foundational understanding in a given field and increases the ability to cooperate internationally. However, some standards that have been developed to meet the requirements of a multinational environment may not be an exact fit for the national needs of smaller nations. While the standard may not be an exact fit there are opportunities to collaborate with partners within the Defence Standardisation Community to learn from their experience and develop the best practice standards in the management and maintenance of standardisation tailored to national needs. However, there is a threat that caused to standardisations credibility as there are typically small staff sizes dedicated to military doctrines in smaller countries which will cause challenges to managing and maintaining up to date doctrines (Nisser, 2021). From this analysis it can be seen that the strengths that standardisation provide in readymade solutions far outweigh the challenges posed by the requirement to adapt them to a smaller force.

Organisation.

Standardisation provides a small defence force with access to the most up to date thought on military organisational concepts. These concepts are flexible enough to allow them to be tailored to meet the needs of smaller organisations (NATO, 2023b). However, while there is some flexibility for tailoring structures of general capabilities, there is likely to be less flexibility in respect of niche capabilities that require logistical tails much larger than the primary capability. Organisations can optimise their force design through collaboration with partners and the application of best practice standards, the result of international agreement and experience (NATO, 2025b). While noting the potential threat to the organisation posed by the difficulty in maintaining small specialised units. From this analysis it is clear that the benefits that standardisation provide to delivering general capabilities outweighs the challenges posed by niche capabilities.

Training.

Standardisation provides a means to evaluate national capabilities through training and evaluation programmes based on best practice standards. It provides a bench mark standard for small military forces to attain (Danish Standards Foundation, 2015). However, this can be limited by the capacity of a military to conduct large enough scale exercises to effectively evaluate the capability. This can be somewhat mitigated through the coordination with partners

to deliver externally validated certification of military capability through cooperative evaluation of training within programmes such as the Operational Capability Concept (OCC) Evaluation and Feedback Programme (NATO, 2022c). There may be difficulty in ensuring that training is keeping pace with technological advancements, such as with UAV and AI, which may challenge standardisations reputation and credibility. From this analysis, the opportunity that standardisation provides to coordinate with partners for external validation outweighs the threat of keeping pace with technological advancements.

Materiel.

Adoption of defence standards in the materiel field will lead to increased interoperability and reduced costs for procurement and logistical support, especially when operating in a multinational environment (EDA, 2007; Fiott, 2014). It should be noted though that as some standards improve materiel there is a risk that cost may increase within certain areas, particularly in the area of safety. Incorporating standardisation into capability development and procurement planning will provide planners access to off-the-shelf standardised equipment designed to meet the intended capability that a standard lays out (Langlois, 2019). While noting that due to limited national military industry we will be reliant on foreign suppliers. This analysis shows that adopting defence standards increases interoperability and reduces procurement and logistical costs, especially in multinational environments. These benefits outweigh potential cost increases in specific areas like safety.

Leadership.

Standardisation provides agreed international standards within concepts and military thought that can form the foundation of the leadership's PME. An example of this being the Comprehensive Operational Planning Directive, which provides a universal planning language for use in a multinational environment (NATO, 2010). However, organisational human resources matters may result in a limited pool of experienced leaders and instructors making swift adoption of standards a slow and often embattled process as some personnel are slow to accept change. This can be managed in cooperation with partners through International PME exchange programs and MTTs to enhance national understanding (NATO, 2018a). This is not without its challenges though as the private sector are likely to target well trained personnel, whose qualifications are aligned to international best practice. This analysis demonstrates that the strengths of PME coordinated with our international partners in coordination with the

opportunities provided for within MTTs far outweigh any human resources challenges or slow adoption of standards.

Personnel.

Standardisation provides access to knowledge and understanding in respect of capability skill requirements and the level of skill required to be effective, creating a skilled staff (O'Neill & Hinton, 2023). That may in turn then create challenges in retention and specialisation as staff become too specialised or look outside the organisation. While that risk may exist the training of personnel to internationally recognised skill levels will allow them and the organisation to interact and be interoperable with multinational forces headquarters. While this will likely provide a net benefit the organisation must be aware of competing with private and public sector for talent (Martin, 2023). The analysis shows that we must take care when conducting standardisation as the threats posed by talent poaching from external actors outweigh the benefits of interoperability within multinational headquarters.

Facilities.

Aligning infrastructure to best practice standards will allow smoother logistical cooperation particularly when cooperating with multinational forces. For example the utilisation of the standard related to the standard NATO pallet will allow the defence forces to interact more effectively with our European partners and facilitate efficient logistical supply by removing barriers than non-standard pallets may create (NATO, 2022d). However, a small defence force is not likely able to support a large number of specialised facilities, such as logistical bases, due to budget constraints. This can be mitigated through the development of shared facilities with civilian sectors. This will reduce complexity and cost while increasing cooperation nationally to facilitate international cooperation (Sander, 2024). But we must remain aware that there would be a vulnerability to emerging threats, as threats that target one nation implementing a standard will likely target all. From the analysis it can be observed that developing shared facilities with civilian sectors to mitigate budget constraints outweighs the vulnerability to emerging threats targeting standardised systems.

Interoperability.

The key strength of Standardisation is its ability to enhance cooperation through shared uniform practices and process that increase the potential for effective cooperation (NATO, 2009).

However, there may be challenges with potential incompatibility with existing non-standard systems (Cargill, 2011). This may likely impact the credibility of standardisation as personnel try to operate within an imperfect environment. While this may prove challenging in the initial stages of standardisation implementation the real opportunity comes from the increase capability that the organisation would gain in respect of participation in international peace & security missions with our European partners. From this analysis it can be understood that enhanced cooperation through shared practices outweighs potential incompatibility with existing non-standard systems, ultimately increasing capability for international cooperation within operations in support of international peace and security.

Standardisation across the spectrum of DOTMLPF-I offers significant advantages for smaller defence forces, despite some challenges. The benefits of ready-made doctrinal solutions, access to up-to-date organisational concepts, and enhanced interoperability outweigh the limitations of imperfect fit and potential cost increases in specific areas. Opportunities for external validation of capabilities and collaboration with partners provide valuable learning experiences and improve best practices. While personnel retention and competition with the private sector pose threats, the overall gains in international cooperation and mission readiness are substantial. Shared facilities with civilian sectors help mitigate budget constraints, and the increased capability for participation in international missions is a key strength. Although some incompatibility issues may arise initially, the long-term benefits of standardisation in fostering uniform and universal practices that enhance cooperation across all military domains make it a valuable strategy for smaller defence forces seeking to maximise their effectiveness and international integration.

1.5. Summary.

In conclusion, defence standardisation involves creating and implementing guidelines to enhance communication, evaluation, and consistency across fields. Its primary aims include reducing complexity, increasing efficiency and quality, improving communication, and fostering cooperation. In a military context, this standardisation enables smooth cooperation among multinational forces, enhancing interoperability, improving resource management, and boosting operational effectiveness. Key goals include enhancing interoperability, improving resource management, and boosting operational effectiveness, providing a uniform means to communicate and cooperate in an unpredictable global environment, which helps build national

military capabilities. Standardisation has been a fundamental aspect of human civilisation, evolving from early engineering and commercial practices to modern military and industrial applications.

Its development can be traced through four key periods: early origins, post-industrialisation, post-World Wars, and the modern era. Throughout history, standardisation has addressed challenges of mass production, interoperability, and cost reduction. In the military context, it has been crucial for equipping armies, enhancing logistics, and promoting cooperation. The post-World War period saw the establishment of international standardisation bodies, while the modern era has focused on tactical doctrines and digital accessibility. Ultimately, standardisation has proven to be an effective solution for simplifying practices, improving cooperation, and reducing costs in complex organisations, particularly in resource-limited military environments

Defence standardisation offers significant benefits for smaller defence forces, providing ready-made doctrinal solutions, organisational concepts and training benchmarks developed by international experts. While implementing standards may present challenges such as adaptation costs, limited personnel and potential incompatibility with existing systems, the advantages outweigh these drawbacks. Standardisation enhances interoperability with multinational forces, reduces procurement and logistical costs, and enables effective participation in international peace and security missions. It also provides access to professional military education and facilitates cooperation through shared practices. Despite concerns about maintaining specialised units and keeping pace with technological advancements, standardisation is a valuable tool for smaller military organisations to exploit.

CHAPTER TWO: CASE STUDY: THE DEFENCE STANDARDISATION LANDSCAPE WITHIN THE DEFENCE FORCES.

2.1. Introduction.

It is without doubt that the Defence Forces apply standards on an almost daily basis. However, in order to establish a baseline of where the Irish Defence Forces is on its standardisation journey, this chapter will examine the current defence standardisation landscape through a SWOT² analysis of a DOTmLPF-P³ framework (US DoD, 2021). This framework provides a broad look at the various components that make up the current defence standardisation ecosystem, it allows for a balanced analysis of all the strengths of the current system, the weaknesses of this system, the opportunities available through cooperation with partners and potential threats that exist within the current system. The DOTmLPF-P framework provides an effective means to establish how each of these components affect the overall system and how they interact and shape each other. Throughout this case study, I will also attempt to establish whether standards have been adopted in a de facto or a de jure manner. A de facto standard is one that has been introduced in practice but has not been accepted through an official process, a de jure standard is one which has been introduced officially, usually through a standardisation development organisation (Campbell, 2012). Developing this understanding of whether, how and why any standards have been adopted within the various Services of the Defence Forces will be important in shaping an understanding of the overall current standardisation environment.

2.2. Evaluating Current Defence Forces Standardisation.

2.2.1. Policy.

Policy is the direction that drives what organisations do. Due to its importance as a driver this is the logical starting point. Standardisation was first identified within policy through the 2015 White Paper on Defence, which identified NATO as the standard to which the organisation will pace itself (Department of Defence, 2015). This was then complimented

² SWOT - Strengths, weaknesses, opportunities and threats.

³ DOTmLPF-P - Doctrine, Organisation, Training, Materiel, Leadership, Personnel, Facilities and Policy.

by the Defence Forces Standardisation Strategy 2015 which represented a positive step for the adoption of standards within the Defence Forces (Óglaigh na hÉireann, 2015a).

Further efforts led to a new standardisation partnership goal being agreed between Ireland and NATO within the Individually Tailored Partnership Programme (ITPP) (NATO, 2024b). This partnership outlined milestones which sought to develop understanding, governance structures, governance mechanisms and standardisation training and education. While this all represents positive progress, adoption of NATO standards within the Defence Forces has been slow. As of the time of writing, the Irish Defence Forces has officially adopted 12 NATO standards (NATO, 2025c). This, however, does not reflect the true level of NATO standardisation within the Defence Forces. For example, a number of NATO publications are issued to students on the Joint and Land Command and Staff Courses as key course reading material. These standards are not currently listed as being adopted by the Irish Defence Forces. This may in part be as a consequence of the current standard adoption process not being agile enough, due to excessive bureaucratic processes, which fail to effectively outline exactly how standardisation will be achieved. The standard adoption process outlined within the strategy also establishes a number of internal organisational standardisation governance bodies such as Management Committee for Standardisation, a High Level Committee for Standardisation, Standardisation Working Groups and Functional Working Groups, however, there are no records available to indicate that these committees or groups ever met or issued direction on standardisation objectives (Óglaigh na hÉireann, 2015). An indication that the governance structures and mechanism outlined within the strategy were not strong enough.

The gaps in the Defence Forces current standardisation strategy can be addressed through deeper cooperation with partnership programmes such as NATO PfP's ITPP. Increasing the Defence Forces participation in NATO and EDA Standardisation Committees and Management Groups will also increase organisational understanding of standardisation and implementation best practices. Recent geopolitical shifts around NATO and the Trans-Atlantic Relationship may create a challenging standardisation in the future as the US pivots away from Europe potentially leading to standard divergence. In short standardisation is effectively outlined in current policy, however, its application demonstrates a de facto application of standards. Efforts are needed to improve its

implementation through revitalised, agile governance creating a de jure application of standards.

2.2.2. Doctrine.

The Defence Forces has recently reviewed its capstone doctrine publication, formerly DFDM-1, now IRLJP-01. This publication was influenced heavily by Allied Joint Publication (AJP) – 01, Allied Joint Operations Doctrine (Óglaigh na hÉireann, 2024b, p 9). In addition to this, the publication demonstrates a future doctrinal architecture which is strongly influenced by the model used by NATO; in this case within Allied Administrative Publication (AAP) 47, Allied Joint Doctrinal Development (NATO, 2019). The IRL publication designation and its proposed structure are a strong doctrinal foundation to develop and build from. There are some weaknesses within the organisations understanding of how defence standardisation work. This has led to slow and inconsistent adoption of NATO standards across all services (NATO, 2025c). In addition to this some, frustrated by the agility of the current standardisation adoption process, adopt NATO standards unofficially (O'Regan, 2025). This leads to different services and corps operating off different standards and practices, leading to reduced interoperability.

A further issue with current standardisation is the lack of governance mechanisms to audit standards which are currently within circulation across the Defence Forces. This has created a situation whereby different schools within the military college are working with different versions and editions of NATO standards. The best example being Allied Tactical Publication (ATP) 3.2.1.1 Conduct of Land Tactical Activities. Within the military college there are at least two versions in circulation. The Infantry School adopted ATP-3.2.1.1, version C, edition 1 as IRLJP-3.2.1.1 in May 2024, however, the ATP-3.2.1.1 issued by the Command and Staff School is a version for 2002. There are a number of potential opportunities to resolve these issues. At the most basic level increasing awareness and access to the NSO website, which offers access in a public and protected format for partners, will allow key standardisation stakeholders to understand what current standards are in promulgation. However, an internal standalone standardisation office within the Defence Forces, in line with best European practice, would provide a single touching point between multiple external standardisation bodies

and internal standardisation stakeholders as well as an auditing function to eliminate the circulation of withdrawn standards. Increased Defence Forces participation in NATO capability working groups would also enhance organisation understanding of the direction our partners are going within doctrinal thought. Taking this path may potentially focus the Defence Forces too much towards a European way of thinking only at the expense of the Irish perspective. However, the risk posed by such doctrinal standardisation is outweighed by the benefits of the strong doctrinal foundation which it offers.

2.2.3. Organisation.

The best example of the strengths of current Defence Forces organisational standardisation strengths lie in its participation with the NATO Operational Capabilities Concept (OCC) Evaluation and Feedback Programme. This programme seeks to train, develop and evaluate partner military capabilities to NATO standards (NATO, 2022c) and set capability codes and statements (NATO, 2023b). The Defence Forces has participated in this programme since 2017, successfully training, developing and evaluating four sub-unit sized military capabilities to NATO standard. The level achieved within these evaluations demonstrate that the Defence Forces can deliver interoperable sub-unit military capabilities within a larger NATO/EU led operations in support of international peace and security. While this has been broadly successful, a pattern has developed whereby sub units are formed up, trained, evaluated and then stood down. So while interoperability is reached by the evaluated sub-unit it is then undermined by the stand down and dispersal of personnel and equipment from this unit. Preparation for OCC participation can also be hindered by the lack of a Defence Forces standardisation office in DFHQ. Previous sub-unit commanders under evaluation have raised the issue that the collation of the most up to date standards can be challenging without a national database of standards (Murphy, 2025).

While these issues can be frustrating they are not insurmountable. The introduction of the NATO Capability Codes and Capability Statements offers the Defence Forces a way to build military capabilities that are aligned with our partners. This can be enhanced by the development of a national database of standards which are organised and linked to the capability codes and statements document requirements. There are also opportunities

to broaden the Defence Forces' understanding of organisational standards through cooperation at a national and international level. Developing national liaison with the National Standardisation Authority of Ireland (NSAI) and international liaison through NATO and the EDA would assist in broadening organisational standard sources too. Recent international developments within International Relations may create challenges to current NATO partnership programmes. The loss of access to the OCC programme would remove a key tool of standardisation evaluation that would negatively impact on the Defence Forces ability to certify its interoperability status. The Defence Forces organisational standardisation is broadly in line with NATO standards and there have been some successes in recent years. Broadening liaison with international and national standardisation bodies will offer opportunities for the Defence Forces to develop effectively and in line with best practice.

2.2.4. Training.

The Defence Forces has engaged positively with standardisation training and education during the last number of years. This has been conducted in cooperation with partners such as the EDA and NATO. Beginning in 2023 the Defence Forces provided a student on the Pilot European Defence Standardisation course, this was followed by the provision of a number of students, from across the various services, on the NATO Standardisation Orientation course. This marked a positive step towards the development of standardisation subject matter experts, who can build standardisation, within the Defence Forces. However, while this is a positive step there is still no national defence standardisation training and education programme. The value gained from participation in course provided by partner organisations will not be fully achieved without the Defence Forces following this up with a national training course. This course needs to outline standardisation, its management and processes from a Defence Force perspective. Thankfully there are a number of opportunities to engage with partners to develop a training and education plan for standardisation PME. One such opportunity is provided by the NATO PfP ITPP. The development of a milestone within the Defence Forces standardisation partnership goal which focuses on the development of a clear line of progression for standardisation SMEs as well as the development of a national standardisation course would mitigate any current weaknesses. There are also further opportunities within the broader Defence Forces PME environment to incorporate

standards in a more effective manner, such as the development of a standards database, managed by a Defence Forces Standardisation Office, linking the required standards to the required PME course and training establishment. The development of these training and education opportunities do not come without cost. In the process of conducting this training they will interact with fellow standardisation PMEs from across Europe and North America. There is a threat as a result of the training and networking that they would pursue further career opportunities within the EDA, European Union Military Staff or within other standardisation bodies. The damage to Defence Forces standardisation understanding caused by this threat would be amplified by the small number of SMEs likely to make up any future standardisation cadre. Despite this the potential benefit to the Defence Forces provided by the development of standardisation training and education outweigh the potential threat of staff exiting the organisation to explore external standardisation employment opportunities.

2.2.5. Materiel.

Materiel standardisation within the Irish Defence Forces is interesting as most if not all materiel systems procured and operated by the Defence Forces are de facto NATO standard. Examples of this include the procurement of small arms systems and ammunition aligned with Allied Engineering Publication 97, which outlines the technical performance specifications for NATO small arms ammunition (NATO, 2020a). This is likely as a result of industrial convergence to the standards expected of the majority of European militaries and our geographic location. However, it is a very strong foundation from which to build interoperable capabilities. This has recently been complimented through Ireland adopting the NATO codification system (NATO, 2025d). This will allow for increased interoperability with European partners through a common language and common data base description of inventory items (NATO, 2023c). These strengths make materiel standardisation the most aligned aspect of the current defence standardisation landscape. While the level of standardisation of materiel is better than almost every other element of DOTmLPP-P under analysis in this case study, it is not without fault. The standardisation up to this point has been conducted in a manner to acquire an item of materiel as opposed to a capability. This is best observed in the Equipment Development Plan, which focuses on the procurement of equipment over the development of capabilities across DOTMLPF-I. The establishment of the Capability Development Unit

in 2024 will mitigate this through the implementation on a 12 year plan with 4 year cycles (Mulcahy, 2025). The broad adoption of best practice defence standards offer a different approach to procurement, opening up the area of Military off the shelf solutions (MOTS) (Harris, 2024). MOTS is a military focused subset of the idea of Commercial off the shelf (COTS) concept, which focuses on readily available, mature technology that can be produced by the commercial sector. This can potentially create swifter acquisition times and cost reduction on materiel projects. As the global security environment becomes increasingly unpredictable and challenging the potential for global supply chain disruption increases. The threat of materiel shortages increases in response. This may require a shift from a Just in Time (JIT) to a Just in Case (JIC) model (Phogat, 2022) and a resultant change in attitudes in respect of materiel stock piling. With all of this considered, however, materiel standardisation remains the most aligned aspect of the current defence standardisation landscape.

2.2.6. Leadership.

There are good examples of senior leadership within the Defence Force supporting standardisation as an enabler. The first example of this is that standardisation is included as a regular briefing point at the doctrinal committee, chaired by the ACOS (O'Regan, 2025). This has resulted in a number of NATO standards being adopted into Irish national doctrine (NATO, 2025c). A further example of senior leadership support of standardisation is the inclusion of a standardisation partnership goal within the ITPP 2024-2028 (NATO, 2024b), this goal involved developing senior leadership's standardisation knowledge which was conducted through a briefing to the Defence Board, Service Commanders and Corps Directors during 2024 (Mulcahy, 2025). Finally the adoption of the Bi-SC NATO capability codes and capability statements by the Defence Forces teams conducting planning for future force design and capability development demonstrate that they understand and support standardisation as a key enabler when developing interoperable military capabilities (Mulcahy, 2025).

Despite this positive progress and support, some standardisation challenges persist within the organisation. These challenges are most notable during efforts to adopt NATO standards within the Defence Forces. Despite defence policy identifying NATO standards as the benchmark for the Defence Forces, actual adoption of standards through

the doctrinal and standardisation processes has been slow with only a small number of standards adopted at this time. This can be observed by the low number of standards adopted by the Defence Forces on the NSO webpage (NATO, 2025c). This is likely as a result of the Defence Force joining the defence standardisation late and senior leadership's lack of exposure to it up until recently (Mulcahy, 2025).

To address the challenges that the Defence Forces faces within standardisation senior leadership can seek to increase coordination and cooperation with regional partners through appropriate EDA (EDA, 2024a) or NATO forums (NATO, 2025b). Learning lessons on standardisation implementation from partner nations, through participation within these forums, can aid in improving the standardisation implementation and management models applied nationally by the Defence Forces. This provides the Defence Forces with an opportunity to build a national system based on best practice but it must be based on clear measurements of performance and measures of effectiveness incorporated into national defence planning key performance indicators. Senior leadership of the Defence Forces are likely to face a number of threats in respect of defence standardisation. These threats are likely to focus on the political and public opposition (Cotty, 2023) concern and resource constraints (McNamara, 2022). The matter of political public opposition can be mitigated by ensuring that the Defence Forces remains aligned with national defence policy (Mulcahy, 2025). Irish defence policy currently outlines NATO as the benchmark standard for capabilities to attain (Department of Defence, 2015). Setting standardisation within policy in this way ensures alignment with political intent. The implications of resource constraints is also likely to shape senior leaders decision making and prioritising of standardisation, however, recent polls, likely shaped by global events, indicate that the Irish public are in favour of an increase in defence spending (Leahy, 2025).

2.2.7. Personnel.

Sticking with the idea that NATO standards have become the de facto standards within the Defence Forces, we can also see that this has taken place within the PME environment. This can be best seen through the implementation of NATO's Comprehensive Operational Planning Directive (COPD) within the Joint Command and Staff Course's Operational Campaign Studies. This provides a strong base of knowledge

for the senior leadership of the organisation to conduct planning in a manner that is interoperable with our partners and allows Defence Forces personnel to integrate into multinational headquarters.

While Defence Forces personnel demonstrate an ability to implement defence standards, there are challenges in developing and maintaining sufficient number of qualified standardisation managers across the organisation (O'Regan, 2025). This is in part due to the existing standardisation strategy not effectively allocating responsibility for standardisation with any particular appointment (Óglaigh na hÉireann, 2015a) but also due to staff turnover, with key staff appointments typically being two years in duration, causing standardisation knowledge to be eroded and lost after successive staff rotations. These challenges cause inconsistent engagement with standardisation bodies and limits the benefits that it could potentially provide to the organisation.

To counter these challenges the Defence Forces can take advantage of its NATO PfP partnership status. This membership provides access to the NATO Tasking Authority (TA) known as the NATO Standardisation committee as well as its associated NATO MC Delegated Tasking Authorities (DTA) and working groups (NATO, 2025b). These committees and working groups provide access to and support from leading defence standardisation experts from across NATO members and partners. This can be also be supplemented through increased and consistent cooperation within the mirrored EDA Committee for Standardisation which provides another perspective on standardisation, broadening the Defence Forces experience and knowledge in respect of defence standardisation (EDA, 2024a). While participation within these groups has increased in recent times, including the EDA conducting a Standardisation Management Group meeting in Dublin in May 2024, it must be consistent to maximise its benefits.

Similarly to the weakness, the points raised here are similar to those identified within the training section of this analysis. This serves to demonstrate the link between the training provided and the skills then developed by personnel. These risks centre upon personnel trained in standardisation seeking employment outside the Defence Forces within external standardisation bodies. This all serves to demonstrate that the Defence Forces personnel have the right skills to implement standardisation when given the time and

space to do so, but that the benefits of standardisation are not maximised due to inconsistent engagement.

2.2.8. Facilities.

While the vast amount of Defence Forces facilities have been inherited from pre-independence and, as such, have not been developed to a modern defence standards. However, there are some excellent examples of where standards have been applied in the development of certain infrastructure. This includes infrastructure linked to previously identified, de facto standardised, materiel such as small arms. The Ordnance Corps for example have recognised Joint Service Publication (JSP) 482 Ministry of Defence Explosives Regulation and its successor Defence Safety Authority (DSA) 03 Part 2, within their Ordnance Instructions (Óglaigh na hÉireann, 2024a). This played a role in the specification of the upgrade of Ammunition Storage facilities within the Defence Forces. The Engineer Corps also have adopted JSP documents within their instructions, for example JSP 403 Handbook of Defence Ranges Safety which is referenced within Defence Forces Training Instruction TI 004/2015 Small Arms Ranges (Óglaigh na hÉireann, 2015b). This appears to be as a result of conducting training courses in the UK. Utilising the JSP documents within these areas ensures that they are aligned with a tested and recognised best practice standard. Now while this has developed a strong standardisation base within facilities standardisation it also creates a potential weakness. JSP documents are a national standard, developed and managed by UK DStan⁴, and are influenced by British needs and regulation (UK MoD, 2024). Access to these documents have been sought by interested stakeholders through open source means and not through a designated standardisation office. This creates an issue for higher HQ understanding what standards are being applied across the organisation as there is currently no central registry of such. It also creates a further issue in respect of access. In recent times UK DStan publications are moving behind a firewall and open source access to updated publications is becoming increasingly difficult to impossible.

⁴ DStan is a British body responsible for the development of MOD standardisation policy and management of the portfolio of standards.

These challenges can be mitigated through the development of cooperation agreements with the UK DStan office via the Ireland – UK Staff talks. This cooperation should focus on the development of defence standards exchange and discussion between the two states, particularly within areas of mutual interest. This cooperation, however, is not without threat. DStan publications are based on British regulation. Following BREXIT there has been some diversion of British Standards from European Standards (Reland, 2024). As a consequence adherence to British National Standards alone, without comparing these standards to Regional or International standards such as EU or NATO standards, could create a situation whereby the standards that are applied are no longer aligned with Irish and EU regulation. As such there is a strong defence standardisation within facilities standards, particularly in areas such as weapons and ammunition storage and associated weapons ranges. Any weaknesses and threats can be mitigated through effective governance and standards management.

2.3. Summary.

From the analysis above we can observe that the Defence Forces demonstrates a commitment to standardisation, with a focus towards NATO standardisation, however, effective implementation has been slow. Which policy and leadership understand and reflect its importance, the bureaucratic processes, lack of resources and personnel turn over impact progress. The main challenge, however, is the organisation's "de facto" approach over a more structured "de jure" approach. This approach fragments standardisation and creates confusion as to what standards have been adopted or implemented across the organisation.

The Defence Forces has a number of strengths when it comes to standardisation. Standardisation has been identified in defence policy as a key driver of capability and a benchmark for the organisation to aspire. Standardisation knowledge and understanding within senior leadership has also grown since the introduction of the strategy and now NATO capability codes and statements are utilised as a planning tool. Materiel standardisation is also an area where the Defence Forces has effectively implemented defence standards, with many weapons and weapon systems meeting NATO standards. Finally the inclusion of NATO standards within PME provides future leaders with knowledge of standardisation and its potential benefits across all aspects of the organisation.

While this is a good start, the Defence Forces has some challenges in the way that hinder standardisation. The first of which are the implementation gaps created by a lack of clear standardisation goals within the standardisation strategy. This lack of clear standardisation goals has meant that the adoption and implementation of standards has been ineffective and inefficient. This is compounded by weak governance and a lack of effective leadership responsibility for standardisation, which has hindered effective management and dissemination of standards within the organisation. Finally, the Defence Forces has engaged with the main regional standardisation bodies, NATO and the EDA in an inconsistent and intermittent manner leading to a reduction in general understanding of standardisation within the organisation.

Having identified these challenges the next chapter will address potential ways and means to address them, utilising lessons learned and best practice from regional partners and standardisation organisations.

CHAPTER THREE: LEARNING FROM EUROPEAN BEST PRACTICE DEFENCE STANDARDISATION MANAGEMENT.

3.1. Introduction.

The previous chapter examined the current standardisation landscape within the Defence Forces, through the identification of standardisation strengths, weaknesses, opportunities and threats. This chapter will now take the gaps in the current Defence Forces standardisation that were identified through this process and identify potential solutions to these differences. The core message of this chapter is that we can identify solutions to these gaps through engagement with European partners, learning from their experiences and their best practices.

The three key gaps within Defence Forces standardisation within Chapter 2 are weak governance, inefficient and ineffective implementation and finally inconsistent and intermittent engagement with standardisation stakeholders. This chapter will now examine potential solutions to this disparities in three sections. Section 1 will examine how the Defence Forces can build effective governance. It will address the building of structures and the enshrining of standardisation direction within policy. Section 2 will examine the development of effective implementation of standards. It will do this by addressing the issue of “de facto” versus “de jure” standardisation outlined within Chapter 2 and will also examine the use of databases and other tools to manage standardisation. Finally Section 3 will examine the enhancement of standardisation engagement with regional standardisation bodies. It will do this through an examination of the current European Standardisation Community and how partners engage with these bodies. It will then examine how enhanced engagement with this community can benefit the Defence Forces understanding of standardisation and assist in improving standardisation and interoperability.

3.2. Building Effective Governance.

This section will examine methods to build effective standardisation governance. It will focus on two key areas; governance structures and standardisation direction within a Defence Forces standardisation strategy. It will examine best practice within partner nations and suggest potential solutions to the gaps within the Defence Forces current standardisation landscape.

3.2.1. Structures.

To understand how to build structures we must first examine how regional partners approach standardisation within their force structures. Almost all European states have a permanent unit with responsibility for standardisation within their militaries. To allow for a comparative analysis of partner nations approaches, of various sizes and complexities, this section will focus on the French, German and Danish force structure models.

The French Ministry of the Armed Forces manages standardisation through the permanently structured Defence Interoperability and Standardisation Support Agency (Known in France as A2IND). This agency (*See Figure 3*) is led by the military but it embraces civilianisation through the employment of civilians within its structures. This civilianisation was established to maintain experience and continuity within the agency (Laboulay, 2024).

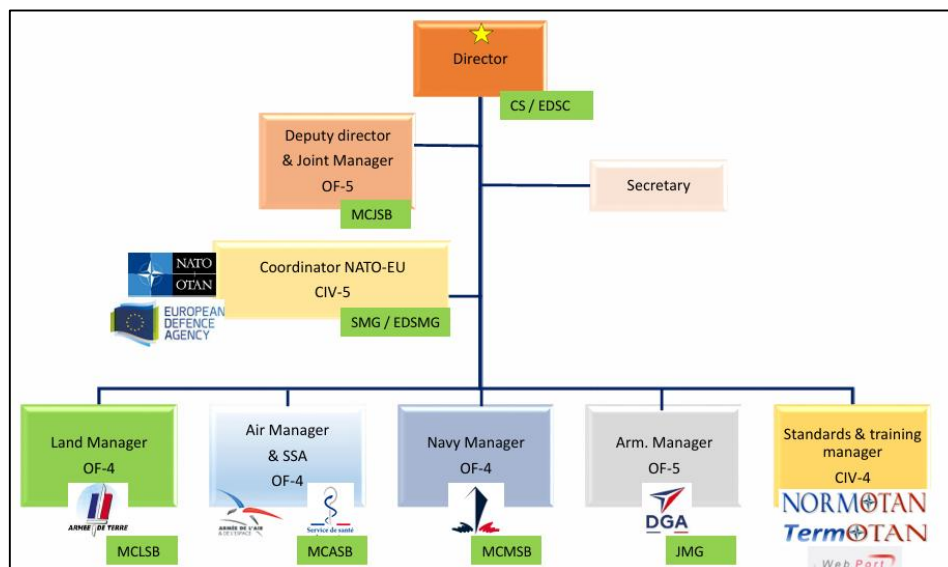


Figure 3: Organisational Structure of A2IND (Laboulay, 2024).

Within this agency there are also representatives from the Army, Navy and Air Force that focus on standardisation specific to their respective service. (Laboulay, 2024). In addition to this the French maintain an Armaments Manager and a Standards and Training Manager (Laboulay, 2024). This agency receives its guidance from a National Standardisation Steering Committee which is joint chaired by the Chief of the Defence

Staff and the Director Defence Procurement Agency (Laboulay, 2024). This model provides a clear structure and direction for standardisation within the French Armed Forces and does so using a balance of military and civilian staff. The civilian staff within the French model overs organisational stability, which can often be a challenge within a military organisation as personnel rotate through appointments regularly. This stability in turn creates increased organisational knowledge and understanding in respect of standardisation plans and processes.

In comparison, Germany takes a different approach. Germany maintains a permanent Standardisation Branch within a Directorate for Scientific Support and Interoperability. This Directorate in turn sits within the Bundeswehr Office for Defence Planning (Weisman, 2023). This model is military led and does not embrace civilianisation in the way that the French model does.

The German Standardisation Branch is subdivided into three sub-branches: Standardization policy, general tasks & training, Central Coordination Center and STAMIS⁵. The branch acts as a central point of contact for the administration of operational, administrative and material⁶ NATO standardization within the Bundeswehr (Weisman, 2023). The German Standardisation Branch coordinates standardisation within the Bundeswehr through nine working groups related to the services and key areas affected by standardisation (*See Figure 4*).

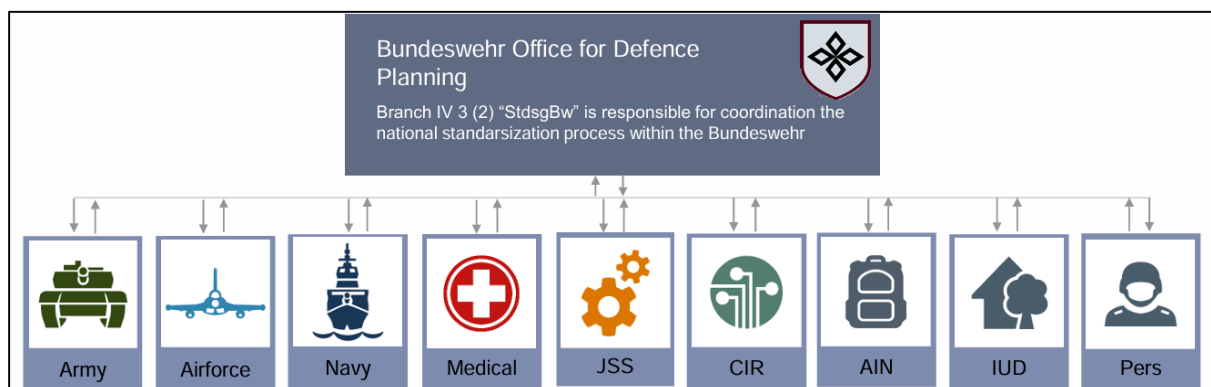


Figure 4: Bundeswehr Standardisation Working Groups (Weisman, 2023).

⁵ STAMIS – Standardisation Management Information Systems. Germany maintains a database of standards accessible across the organisation on this system.

⁶ NATO classify standardisation into three areas operational, administrative and material standards within AAP-03.

These working groups are led by a responsible action officer and supported by subject matter experts (SME). SMEs within these working groups tend to do so as a secondary appointment (Weisman, 2023). In comparison to the French model, the German model is more bureaucratic, stratified and less permanent outside the main branch.

The final model for examination is the Danish model. It is a much simpler model than the other two, however, as you will see, this has a number of draw backs. In comparison to the French and German models, the Danish take a more distributed approach to their standardisation model.

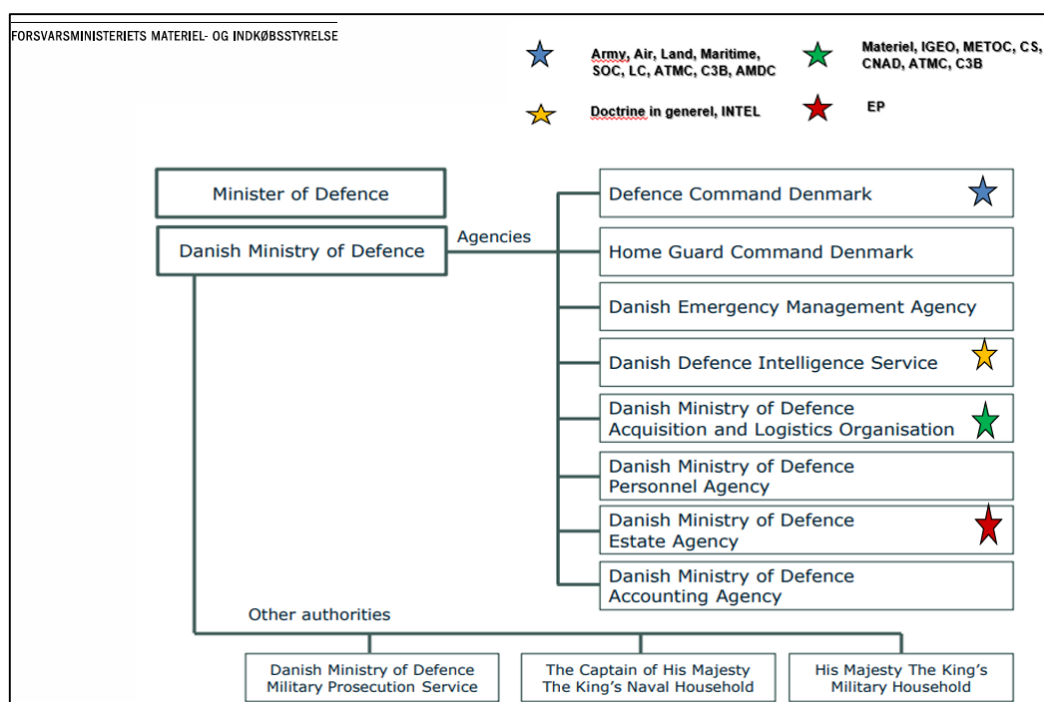


Figure 5: Danish Standardisation Structures (Pederson, 2024).

Unlike the previous approaches the Danish divide operational, administrative and materiel standardisation across a number of services and agencies (See Figure 5). With organisational and administrative standardisation being split across the Defence Command, Intelligence Service and Estate Agency and materiel standardisation being the responsibility of the Defence Acquisition and Logistics Organisation (Pederson, 2024). Ironically, this distributed structure has resulted in an unstandardized approach to standardisation across the Danish Armed Forces, particularly within defence planning (Pederson, 2024). The Danish model also relies on individual appointment

holders across the distributed standardisation structures to keep momentum on standardisation going, this had led to implementation challenges caused by changes in personnel and a lack of SMEs and other specialists (Pederson, 2024).

Of the three structures examined in this section, the French model appears to be the most effective structure due to its permanence, simplicity and mix of both military and civilian staff, a common practice used to maintain standardisation knowledge. This model appears to maintain SME knowledge within the organisation while allowing for effective standardisation governance, overseen and directed by the National Standardisation Steering Committee. By developing a similar structure the Defence Forces could address its current weak standardisation governance through a permanent standardisation governance structure with effective oversight provided by a Standardisation Committee.

3.2.2. Strategy.

To ensure that the structures of an organisation operate effectively they must be first supported by an effective strategy (Chandler, 1962, p. 14). This strategy provides the direction required to effectively allocate resources and achieve the organisation's desired end state. Effective strategies should also contain clear measures of success in the form of goals (Alderstein, 2023). This is achieved within the military in the form of a mission and its supporting objectives, giving a unit a task and purpose (NATO, 2024c, p. 1-6). A mission is a clear, concise statement of the task of the command and its purpose (NATO, 2020, p. 1-1). This provides the direction necessary to allow planners to understand how they will approach their tasks. An example of where this is used effectively is the NATO Strategy for Enhancing Standardisation. This publication outlines a vision that identifies that NATO wishes to enhance standardisation in support of interoperable capabilities for missions and operations in support of Alliance core tasks (NATO, 2018b, p. 1-1). It follows this vision with a clear outline of the ends that it wishes to achieve (interoperability), the ways (processes) that can meet these ends and the means (standards and evaluation) that are available to support the achieving of the both ways and ends (NATO, 2018b, p. 1-1 – 1-3). This publication provides sufficient information to allow standardisation managers and national standardisation

offices to formulate their respective national plans. It also can serve as a solid base from which to develop national standardisation strategies.

In order to measure success the military utilises measures of performance (MOP) and measures of effectiveness (MOE). A MOP is defined as the criteria used to evaluate the accomplishment of actions (NATO, 2010, p. 5-18). An example of a MOP within standardisation is the recording of the number of standards adopted by a nation. Within the NATO Standardisation Process Allies are obliged to report the implementation status of a standard, partners are invited to report on standards adopted (NATO, 2022a, p. 32-33). This reporting process provides a means to measure performance. A MOE is to describe the specific changes in the capabilities of a system or subsystem to be able to establish that desired effects are or are not being achieved (NATO, 2010, p. A-16 – A-17). An example of a potential MOE is the results of an evaluation and feedback programme undertaken by a unit training exercise. The Operational Capabilities Concept Evaluation and Feedback programme (NATO, 2022c), currently undertaken by the Irish Defence Forces is an example of such a programme. This programme aims to improve the interoperability of NATO partner nations through a programme that evaluates partner's military capabilities in accordance with NATO standards (NATO, 2022c, p. I-3 – I-6). The evaluation feedback provides partner nations a metric from which to compare changes in capability based on changes of MOPs, providing an effective MOE. To ensure that an overarching and systemic approach is taken in respect of standardisation it is important to coordinate top down planning goals that are contained within MOP and bottom up feedback provided within MOE to achieve the end state of standardisation, which is interoperability (NATO, 2016, p. 1-1). Combining a mission, MOPs, MOEs within an overarching systemic approach that outlines the ends, ways and means to achieve interoperability will deliver an effective strategy that supports the organisational standardisation structures to operate effectively.

Suitable structures and effective strategies are crucial for achieving an organisation's standardisation goals. The French model provides a suitable model for standardisation governance through its hierarchical structure, maintained expertise and oversight provided through a Committee for Standardisation. This model can serve as a foundation from which to develop a model scaled to the Defence Force's needs. Meanwhile, a comprehensive strategy outlines the vision, mission, and objectives, supported by measures of performance and

effectiveness. Together, these elements ensure coordinated efforts, efficient resource allocation, and a systematic approach to implementing standards. This synergy between structure and strategy is essential for enhancing interoperability and meeting standardisation objectives across military organisations.

3.3. Developing Effective Implementation.

Building on the previous section, this section will examine the development of effective standardisation implementation. It will focus on standardisation implementation planning, standardisation implementation support tools and their links to defence planning. It will examine best practice within partner nations and suggest potential solutions to the gaps within the Defence Forces current standardisation landscape.

3.3.1. Addressing De Facto v De Jure Standardisation.

As discussed in Chapter 2 current standardisation within the Defence Forces relies heavily on “de facto” standardisation which in turn has led to fragmented standardisation across the Services and Corps. This fragmentation in turn creates internal and external interoperability challenges that impact the Defence Forces in a negative way.

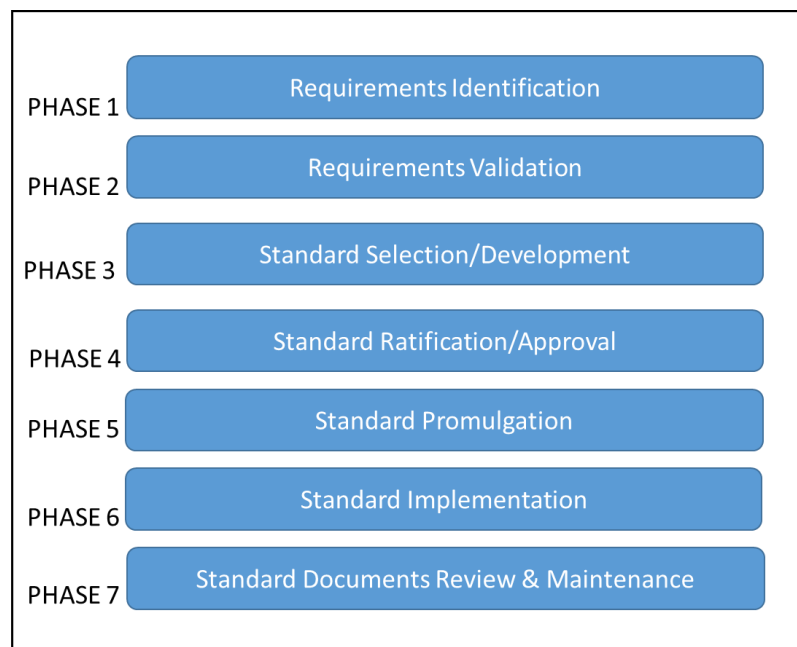


Figure 6: NATO Standardisation Process (NATO, 2022a).

To reduce this fragmented approach to standardisation the Defence Forces must examine means to reduce its reliance on “de facto” standardisation and develop a more balanced and planned approach that favours a “de jure” standardisation. NATO, while a regional organisation, must also manage standardisation not just across one organisation but across many nations. To do this NATO utilise Allied Administrative Publication 03 Directive for the production, maintenance and management of NATO standardisation documents (NATO, 2022a).

This publication establishes procedures for all NATO standardisation stakeholders involved in the production, maintenance and management of standards (NATO, 2022a, p. 5). It lays out a common seven step standardisation process (*See figure 6*) which identifies standardisation gaps, develops standardisation tasks to meet these disparities, selects the correct standards to meet these tasks, followed by an approval, distribution, implementation and review steps (NATO, 2022a, p. 14).

The NATO Standardisation Process is a crucial mechanism for enhancing interoperability and effectiveness among NATO forces. It is closely linked with the NATO Defence Planning Process (NDPP) to ensure that standardisation efforts align with NATO's strategic objectives and capability requirements (NATO, 2021, p. 8). Due to the standardisation processes links with the NDPP it primarily operates a top down standardisation system based on standardisation tasks developed by the relevant TA/DTA (NATO, 2021, p. 11). This ensures a planned and coordinated approach to standardisation. NATO's standardisation process provides a strong baseline that the Defence Forces can adapt and adopt to meet its specific standardisation needs.

In order to effectively implement standards, the Defence Forces must move away from its current fragmented "de facto" standardisation model to a more structured "de jure" approach to enhance interoperability and operational effectiveness. NATO's standardisation process, as outlined in AAP-03, offers a suitable model for implementing a comprehensive, top-down approach to standardisation to address this. By adapting and adopting a similar model, the Defence Forces can develop a tailor made implementation model to address its needs. Moving to a more deliberate and coordinated standardisation implementation approach will mitigate current challenges but also enhance internal

standardisation cohesion within the Defence Forces and external standardisation collaboration with partners.

3.3.2. Standardisation Support Tools.

There are a number of key planning tools to support standardisation planner's efforts. A common trend among Standardisation bodies is the development of a standardisation database. While both NATO and EDA maintain standards databases their approaches differ. Within NATO, the main standardisation tools include the Capability Codes and Capability Statements (CC&CS) document and the NATO Standardisation Document Database (NSDD). The CC&CS provides a common language for describing capabilities in defence and operational planning. The NSDD offers consolidated storage of all NATO standardisation documents. Additionally, the European Defence Agency's EDSTAR complements these NATO tools in promoting interoperability and standardisation. This section will now outline the key benefits of these systems and how they can potentially interact.

NATO operate a database known as the NATO Standardisation Document Database (NSDD). The NSDD is the authoritative source for accessing NATO standardisation documents (NATO, 2018b, p. 1-4). As a tool for supporting standardisation efforts, the NSDD plays an important role in enhancing interoperability and operational effectiveness among NATO allies and partners. It provides the Defence Forces access to current promulgated documents, ensuring that all stakeholders have up-to-date information on NATO standards (NATO, 2022b). This centralised database enables efficient dissemination and management of standardisation documents, with an average of 30,000 standards downloaded monthly from over 90 countries (Akşit, 2014, p. 4). By providing a single source for all NATO standardisation materials, the NSDD supports the NATO Defence Planning Process (NDPP) in achieving interoperability and coordinating the development of forces and capabilities for Allied operations and missions (NATO, 2022b).

The EDA also operate a database known as the European Defence Standards Reference System (EDSTAR), but it takes a different approach. EDSTAR is database containing references to best practice (EDA, 2025a). It has two distinct elements the Expert Group

Reports and the Best Practice Standards. Unlike the NSDD, EDSTAR is not a repository of standards but instead provides a pathway to the best practice standard identified (EDA, 2025a). It currently contains guidance for approximately 2300 standards (EDA, 2022). The standards reference on EDSTAR are global in their span and include both civilian and military standards too (EDA, 2025a). By broadening the source of best practice standards the EDA can assist industry and member states from becoming reliant on a single source (EDA, 2022). In addition to the references to best practice standards EDSTAR also includes an Expert Group Report section. This section contains technical reports by leading SMEs which list best practice standards across 28 technical domains (EDA, 2024a). This tool provides the Defence Forces with access to project research, leading to lower research time and cost (EDA, 2022). EDSTAR demonstrates therefor represents a separate but tool for standardisation stakeholders that can be utilised in conjunction with the NSDD.

A further supporting tool of standardisation are the NATO Capability Codes and Capability Statements. The NATO Capability Codes and Capability Statements are a defence planning tool and product of the NATO Defence Planning Process that link standards to capabilities (NATO, 2023b, p. 1). They provide the traceability required between interoperability requirements and standardisation solutions (NATO, 2021, p 8). It achieves this by referencing the standards required to achieve each capability (NATO, 2023b, p. 11). This planning document is used across all NATO and EUMS operational and defence planning (NATO, 2023b, p. 1) providing a common planning language across the region for militaries and assisting commanders to understand the standards that they require to achieve their respective capabilities.

Standardisation support tools like NATO's NSDD and CC&CS, along with EDA's EDSTAR, play crucial roles in enhancing interoperability and operational effectiveness. These complementary systems provide centralised access to standards, best practices, and capability requirements, facilitating efficient planning, research, and implementation across NATO allies and EU member states. Their combined use by the Defence Force strengthens regional defence cooperation with our partners and broadens the reach of our standardisation efforts.

In summary, the Defence Forces must transition from a fragmented "de facto" standardisation approach to a more structured "de jure" system to enhance interoperability and operational effectiveness. This shift requires adopting a comprehensive top-down approach, similar to NATO's standardisation process, which aligns with strategic objectives and systematically addresses gaps across all services and corps. Key to this transition are standardisation support tools such as NATO's NSDD and CC&CS and EDA's EDSTAR. These complementary systems provide centralised access to standards, best practices, and capability requirements, facilitating efficient planning, research, and implementation. The NSDD offers a single source for NATO standardisation documents, while EDSTAR provides references to global best practices and expert reports. The CC&CS links standards to capabilities, offering a common planning language across the region. By leveraging these tools and adopting a more deliberate and controlled standardisation strategy, the Defence Forces can improve internal cohesion, external collaboration and overall readiness in an increasingly complex military landscape.

3.4. Enhancing Standardisation Engagement.

Finally, this section will examine European Defence Standardisation Organisations. It will focus on NATO and the EDA, particularly their engagement opportunities. It will examine best practice within partner nations and suggest potential solutions to the gaps within the Defence Forces current standardisation landscape.

The two major players within European Defence Standardisation are NATO and the EDA (Laasme, 2010). Both of these organisations operate their own respective governance and supporting structures, however, each organisation takes a different approach to standardisation. The key engagement opportunities within these organisations are the NATO Committee for Standardisation (CS) (NATO, 2020b), the NATO Standardisation Office (NSO) (NATO, 2024a) and the EDAs European Defence Standardisation Committee (EDSC) (EDA, 2020).

NATO and the EDA are two key entities in the realm of European Defence Standardisation. While they share some similarities, their differences reflect the distinct geopolitical contexts in which they operate. The key similarities of the standardisation efforts of both the EDA and NATO are a focus on interoperability among their members, their engagement with standardisation stakeholders and their governance structures that operate under larger frameworks (EDA, 2022, NATO, 2022b). But while they do have these similarities they do have a number of key differences.

The NATO CS is the senior policy committee which is responsible for standardisation policy and management within NATO which reports to the North Atlantic Council (NAC) (NATO, 2025b). In contrast, the European Defence Standardisation Committee (EDSC) functions under the European Defence Agency (EDA), which is part of the EU's Common Security and Defence Policy (CSDP) framework (EDA, 2025b). The scope of standardisation activities between NATO and the EDA also differ. The CS focuses on standardisation across NATO member states and partners to ensure military interoperability (NATO, 2025b). The EDSC, on the other hand, aims at harmonising defence standards specifically across EU member states to strengthen the European Defence Technological and Industrial Base (EDTIB) (EDA, 2024a).

The outputs of both organisations also differ. The CS produces a range of standardisation concepts, doctrines and procedures to achieve and maintain interoperability between Allies and Partners (NATO, 2022b), through the NSO (NATO, 2025c). The CS coordinates all standardisation activities, monitors the implementation of these standards across Allies and Partners and oversees the work of the NSO (NATO, 2020b). The CS participation is open to Allies and Partners and Observers if invited to participate (NATO, 2025b). Subordinate to the CS, NATO formed a Standardisation Management Group (SMG) to assist the CS in achieving its standardisation mission, which is open to NATO Allies and Partners (NATO, 2025b). These standards are disseminated through the NSDD (NATO, 2025a). The EDSC, in comparison, utilises EDSTAR, a reference system for "Best-Practice Standards" in defence projects, which is more focused on identifying and referencing existing standards rather than creating new ones (EDA, 2025a).

EDSC also operates a subordinate European Defence Standardisation Management Group to support its standardisation mission (EDA, 2020b). The CS includes 32 NATO members (NATO, 2025b) and engages with partners through the Interoperability Platform, which is a standing format for cooperation on NATO interoperability issues (NATO, 2024d). The EDSC's membership comprises 27 EU member states, but it also engages with non-EU stakeholders such as Ukraine and Norway, reflecting the EU's broader collaborative approach in defence matters (EDA, 2020). There is extensive crossover of participant states within both of these committees with 23 of the 27 members of the EU being members of NATO (European Council, 2025). NATO's CS and the EDA's EDSC differ in scope, governance, outputs, and membership. With the CS focusing on military interoperability across NATO members and

partners, while the EDSC harmonises defence standards for EU states with a focus on industry. Despite any differences, their overlapping memberships highlight opportunities for collaboration in defence standardisation and provide access to regional standardisation SMEs.

A dual-track approach to participation within the EDSC and the NATO CS can offer significant benefits to the Irish Defence Forces. As such, the Defence Forces should adopt a dual-track approach to engagement with the EDA's EDSC and NATO's CS to leverage their complementary strengths and address their distinct governance structures, scopes, and outputs. NATO's CS focuses on military interoperability across its 32 members and partners, producing comprehensive standardisation doctrines and procedures through the NATO Standardisation Office (NSO). In contrast, the EDSC harmonises defence standards among EU states, emphasizing industrial collaboration and leveraging EDSTAR to reference best-practice standards. While NATO's CS operates under the North Atlantic Council, the EDSC functions within the EU's Common Security and Defence Policy (CSDP) framework, reflecting differing geopolitical contexts. Despite these differences, overlapping memberships—23 EU states are also NATO members—offer opportunities for collaboration and access to regional expertise. A dual-track strategy would enable the Defence Forces to align with NATO's interoperability goals while supporting the EU's focus on strengthening the European Defence Technological and Industrial Base (EDTIB), ensuring comprehensive engagement with both frameworks. This dual track system should be backed up with concrete engagement goals linked to regular attendance of both committees and greater engagement within expert groups by Defence Forces SMEs.

3.5. Summary.

This chapter demonstrated how European best practice can offer effective solutions to the differences identified within current Defence Forces defence standardisation efforts. These gaps include weak governance, ineffective and inefficient implementation and inconsistent engagement with stakeholders. By adopting and adapting the French governance structure module, which combines military and civilian expertise under the oversight of a National Standardisation Steering Committee, the Defence Forces can establish a permanent, robust governance structure that directly addresses the current gap in governance structures. In order to move from the currently fragmented “de facto” approach to a more structure “de jure” approach the Defence Forces can adopt and adapt NATO's seven step process as outlined in

AAP-03. This will enhance implementation through a planned top down strategy. This process is supported by a number of standardisation tools such as the NSDD which offers a central access point for NATO standards, EDSTAR which provides references to best practice standards and the NATO CC&CS which links standards to military capabilities. All of which if used in an integrated way can mitigate implementation inefficiencies. Finally implementing a dual-track engagement strategy with NATO's CS and the EDA's EDSC will enhance collaboration, harness regional expertise and resolve inconsistent stakeholder engagement leading to improved interoperability. By integrating these strategies, the Defence Forces can greatly enhance its operational effectiveness and readiness, in an increasingly complex global security environment, through the adoption of standardised practices that support internal cohesion and external collaboration.

CONCLUSION

This thesis has examined how the Defence Forces can leverage defence standardisation to enhance interoperability with regional partners. It focuses on partnership programmes such as NATO's PfP programme to address the identified standardisation gaps of weak governance, ineffective implementation and inconsistent stakeholder engagement. It proposes a tailor made solution and a way forward for the Defence Forces based on regional best practices. This chapter will synthesis the key findings of the research, provide a reflection on the methodology, discuss areas of further research based on this and will conclude with a general reflection on the research topic.

Through research this thesis has identified a number of key findings. Firstly it established that defence standardisation, as outlined within national and international policy and within historical publications such as Perry (1955) and Verman (1973), is a crucial mechanism for reducing complexity, enhancing efficiency and enhancing interoperability. Secondly it outlines how standardisation can benefit smaller military forces through the opportunity of ready-made doctrinal solutions, access to contemporary organisational concepts and access to international experts. However, analysis through a DOTmLPF-P framework demonstrates that current rates of standardisation adoption are slow, primarily due to weak governance. This thesis proposes a solution to this through the adoption of a permanent standardisation office based on the French model, supported by a strategy that identifies clear MOPs and MOEs linked to interoperability. Thirdly it identified that the Defence Forces currently conducts standardisation from a de facto approach. This appears to be the case, despite policy and senior leadership recognising the importance of standardisation. It would appear that progress is limited by ineffective implementation mechanisms, resource constraints and high staff turnover. This thesis proposes the development of a national standardisation implementation model based on AAP-03, focusing on top down implementation, supported by bottom up recommendations. This model will provide a mechanism to move from a de facto approach to a de jure model. Finally it examined how the Defence Forces has engaged with regional standardisation bodies, identifying that since the development of the Defence Forces Standardisation Strategy 2015 there has been inconsistent engagement. It demonstrates how inconsistent engagement has led limited the organisations standardisation understanding which in turn limits its ability to develop interoperability. This thesis proposes a dual track approach

engaging with both NATO and the EDA to enhance standardisation engagement and build standardisation knowledge and understanding.

The thesis's methodology employed thematic literature reviews, historical analysis and comparative analysis to identify the insights into defence standardisation. By applying a SWOT analysis to the DOTMPF-I and DOTmLPF-P frameworks, a detailed examination of standardisation across the Defence Forces was possible. A limitation of this research is its limited geographic scope. The focus on European partners over global partners narrows the scope of analysis and potentially overlooked standardisation opportunities provided by other sources. This decision, however, was based on defence policies recognition of NATO standards as the benchmark for the Defence Forces. Finally, the research focused on an academic analysis of the problem statement and did not explore the potential cultural resistance to standardisation. This cultural piece could be explored in further research on the topic. These limitations suggest that while the findings of this thesis are comprehensive, they may require further validation through primary research.

The implications of this research extend into the broader topic of military and security studies. It addresses a void in current defence standardisation research, which addresses standardisation within larger states and regional organisations but fails to provide solutions to the standardisation challenges faced by smaller militaries. It offers a practical solution combining a French style governance model, an implementation process based on NATO's seven step process as outlined within AAP-03 and a dual-track engagement approach with NATO's CS and the EDA's EDSC. This provides the Defence Forces, or a similar sized military, which is aligned with defence policy, with a cost effective solution to align itself with regional best practice standards. Standardisation not only offers reduced procurement and logistical costs but also develops and enhances military capabilities operational readiness for military operations conducted in cooperation with regional partners.

Further areas of research into defence standardisation and its impact on the Defence Forces are numerous, however, the following are the most relevant to the findings of the thesis. The first area of research proposed is a pilot project which tests the governance mechanisms and implementation mechanisms proposed within this thesis. This would be most effectively conducted within a unit under OCC evaluation as they will be required to adopt NATO standards in preparation for an interoperability evaluation. The second proposed area of further research would be the expansion of the comparative analysis to include a global scope through

the inclusion of non-European partners. This may highlight broader global trends not identified within the European focus of this thesis. Finally, an examination of the cultural resistance to standardisation within militaries, potentially focusing on but not limited to the Defence Forces, could provide a more holistic understanding of how standards can be effectively implemented. These proposed areas of research recognise the foundation provided by this thesis while outlining potential areas for further development within standardisation that can benefit the Defence Forces.

The intent of this thesis was to highlight the strategic importance of defence standardisation to the Defence Forces. Identifying it as a means to reduce complexity and cost while increasing efficiency and quality, through the development of universal, transparent and uniform means of communication, that remove barriers to cooperation in order to foster interoperability with partners. By addressing weak governance, ineffective implementation and inconsistent engagement the Defence Forces can build interoperable military capabilities that enhance national capacity and international cooperation. In an increasingly unpredictable world, standardisation provides the Defence Forces with a predictable foundation from which to grow.

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